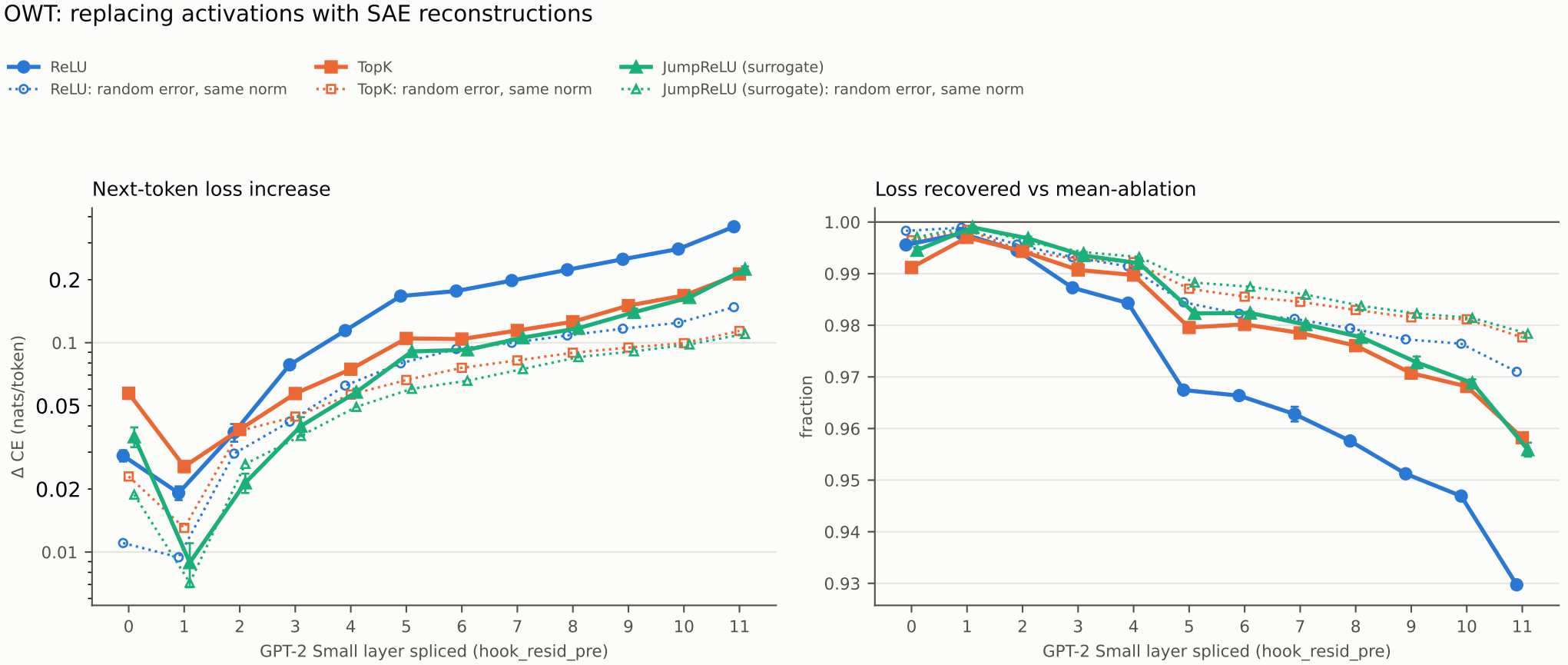

Validation split; mean ± sd over sparsity-matched seeds (42/1042/2042); layer 0 JumpReLU seed 42 excluded. Dotted: the same run with its SAE error replaced by a random error of equal per-token norm.